YONITE EDUCATION INC
新学力

ECON 4140

Class1 公开课

讲师：Patrick

机构介绍

新学力 Yonite，全球领先的教育咨询与海外生活，全方位平台。为每一位学子提供定制化的、全程式的海外留学规划与实现方案，以及职业发展支持。我们致力于激发学生的潜能，帮助他们实现在国际舞台上的成功，同时促进跨文化交流和全球合作。

导师团队

其他产品清单&其他产品介绍

申研产品

加拿大首家按照区域（加拿大，美国，英国，澳洲，香港）组建申研团队的机构
创立 10 年，申研成功率达 100%

求职产品

1. 毕业工作保录/内推
2. 在校实习
3. 面试培训
4. 技能培训

具体可咨询小助手了解详情~

自我介绍

姓名：Patrick

- 满分绩点毕业于约克大学计算机 &统计双学位
- Resident‘s honor roll 成员，奖学金拿到手软
- 8.5 年教学经验
- 每学期带领超 50+学生 A/A+
- 教授大一-大三同时跨领域授课于 **ECON（经济）、MATH（数学）、ADMS（管理）、EECS（计算机）** 等多个学科，实现从大一基础课程到大三进阶课程的**无缝衔接**与系统辅导。
- 熟悉宏观教授的出题套路，让学习更加轻松

课程简介

Econ 4140 Financial Econmetrics

- Prerequisite: AP/ECON 2500, 3210/3500
- 统计进阶课。作为 2500，3210 的延伸，这门课将围绕统计和在金融中的应用展开。Midterm 考察回报率，时间序列模型，final 主要讨论资产定价模型和其他金融模型
- 核心主题：收益与随机游走/EMH、AR(1) 时间序列、投资组合与 Sharpe 比率、CAPM、ARCH 波动模型、期权定价与无套利、Value-at-Risk（VaR）、固定收益证券。
- 理解金融计量核心概念；掌握金融数据性质；能估计基础动态模型并解读计算机输出；提升基础编程与实证能力。

教授介绍

Ji Qi

- 课程结构清晰，讲课节奏稳定
- 考试题型较为常规，重点明确
- Midterm 偏重基础概念与公式理解
- Assignment 包含统计软件应用（SAS/R/Python）
- Final 综合性更强，更注重模型理解与应用

整体而言，只要课堂内容与练习题掌握扎实，获得 A/A+ 是完全可实现的。

课程考试占比

Course Evaluation

Assessment	**Weight**
Group Assignment 1	10%
Group Assignment 2	10%
Midterm Exam	20%
Group Project 3	20%
Final Exam	40%

- Midterm：May 25（课堂时间）

- Final Exam：学校统一安排
- Assignment 包含统计软件分析与结果解释
- 支持 SAS / R / Python 等工具

内容依据 Summer 2026 官方 Outline 更新。

全程班暂时课程安排

Full Course Schedule

Class	**Topics**
Class 1	Review of Probability & Statistics; Financial Returns
Class 2	Time Series Models; AR(1); Assignment Review
Class 3	Portfolio Theory; Sharpe Ratio; Midterm Review
Class 4	CAPM; Asset Pricing Models
Class 5	Volatility Modeling; ARCH; VaR
Class 6	Option Pricing; Final Review

今日课程目标

Lecture 1.1 Probability and Statistics

Lecture 1.2 Return: Price and returns

Lecture 1.1 Probability And Statistics

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2. Probability vs Statistics

Key Concept

Probability	**Statistics**
已知总体 → 推断结果	已知样本 → 推断总体
理论模型	数据分析
Parameter	Estimator

常见符号

Population	**Sample**
μ（总体均值）	x̄（样本均值）
σ²（总体方差）	s²（样本方差）
σ（总体标准差）	s（样本标准差）

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3. Measures of Central Tendency

3.1 Mean（均值）

Population Mean

$$\mu = \frac{\sum x_i}{N}$$

Sample Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Financial Interpretation

Mean return 表示:

资产长期平均收益率

是金融分析中最核心指标之一。

Practice 1

样本数据:

2，4，6，8，10

求：

- Sample Mean
- Population Mean（假设这是总体）

Solution

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6$$

若视为总体：

$$\mu = 6$$

3.2 Median（中位数）

Key Concept

中位数：

排序后位于中间位置的数据

Important Property

Median 对异常值（Outliers）不敏感。

因此：

- skewed distribution 更适合使用 median
- symmetric distribution 更适合使用 mean

Practice 2

数据：

1，2，3，4，100

求：

- Mean
- Median

并分析哪个更适合描述数据中心。

Solution

Mean：

$$\bar{x} = 22$$

Median：

3

由于存在极端值 100：

- Mean 被严重拉高
- Median 更稳健（robust）

3.3 Trimmed Mean

Key Concept

Trimmed Mean：

去除两端极端值后计算平均数

用于：

- 减少 outlier 影响
- 保留 mean 的统计性质

4.1 Variance & Standard Deviation

Sample Variance

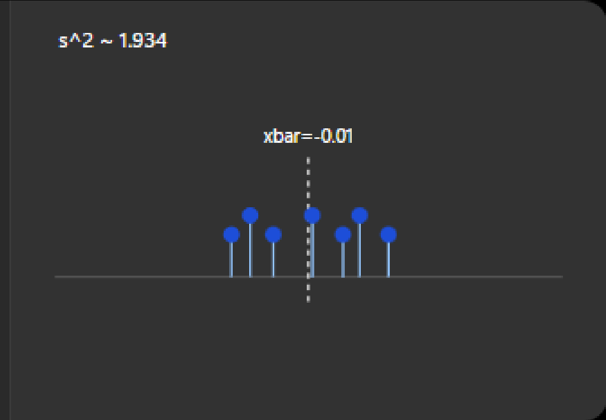
×

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

$spread$ 2.0

n 7.0

$\bar{x} \approx -0.01,; s^2 \approx 1.934$



Standard Deviation

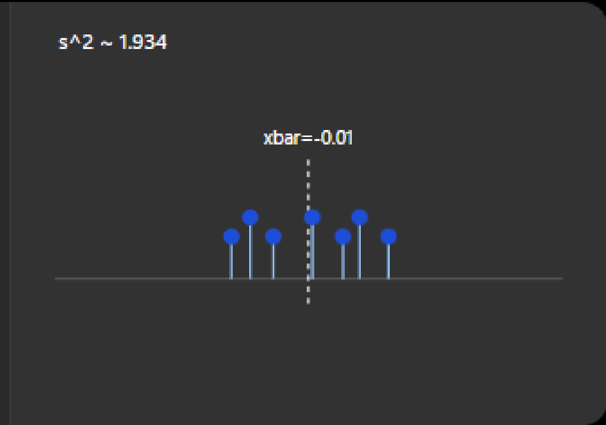
×

$$s = \sqrt{s^2}$$

$spread$ 2.0

n 7.0

$\bar{x} \approx -0.01,; s^2 \approx 1.934$



Financial Interpretation

Standard deviation 在金融中表示:

Volatility（波动率）

标准差越大:

- 风险越高
- 收益波动越大

Practice 3

股票 A:

- Mean Return = 10%
- Standard Deviation = 2%

股票 B:

- Mean Return = 10%
- Standard Deviation = 8%

哪个风险更高?

Solution

股票 B 风险更高。

因为:

$$8\% > 2\%$$

标准差越大:

→ return volatility 越高

4.2 Coefficient of Variation (CV)

Key Formula

$$CV = \frac{s}{\bar{x}}$$

Financial Meaning

CV 衡量:

单位收益对应的风险

适用于:

- 比较不同收益水平资产

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5. Percentiles & Quartiles

Quartiles

Quartile	Percentile
Q1	25%
Q2	50%
Q3	75%

Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

Important

IQR:

- 不容易受 outlier 影响
- 常用于 skewed data

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6. Normal Distribution

Empirical Rule（68-95-99.7 Rule）

对于 Normal Distribution：

Range Approx. Probability

μ ± 1σ 68%

μ ± 2σ 95%

μ ± 3σ 99.7%

Financial Application

用于：

- Risk estimation
- Confidence interval
- VaR analysis

Practice 4

已知：

- Mean return = 5%
- Standard deviation = 2%

假设收益服从 normal distribution。

求：

收益落在：

$$1\% \sim 9\%$$

之间的大致概率。

Solution

$$5\% - 2(2\%) = 1\%$$
$$5\% + 2(2\%) = 9\%$$

即：

$$\mu \pm 2\sigma$$

根据 empirical rule：

$$95\%$$

7. Financial Returns

7.1 Net Return

Formula

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Practice 5

股票价格:

- Yesterday: 100
- Today: 110

求 net return。

Solution

$$R_t = \frac{110 - 100}{100} = 0.10$$

即:

$$10\%$$

7.2 Gross Return

Formula

$$1 + R_t = \frac{P_t}{P_{t-1}}$$

Practice 6

若:

$$R_t = 0.08$$

求 gross return。

Solution

$$1 + R_t = 1.08$$

7.3 Log Return

Formula

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Why Log Return?

金融计量中更偏爱 log return，因为:

- 可加性（additivity）
- 更适合时间序列分析

- 更接近正态分布

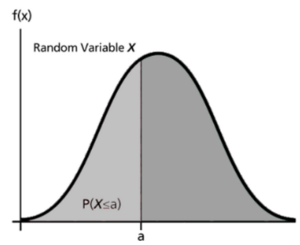
知识点梳理

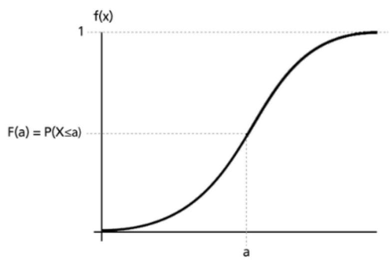
1. Random Variable

1) Large set of possible values
2) Only one will occur
3) The set of possible values and their probabilities = the probability distribution
   - Throw a coin, 2 possible results head, tail with probabilities P(H) = 0.5, P(T) = 0.5

2. Continuous random variable

1) Probability density function (PDFf) f_x: P(X $\in$A) = $\int_A f_x(x)dx$ dx for all sets A
2) Cumulative distribution function (CDF): $F_x(x) = P(X \leq x)$





真题演练

Q1. Find the probability that a standard normal variable takes a value between 0.3 and 0.7.

知识点梳理

3. Quantiles

1) If the CDF of X is continuous and strictly increasing then it has a inverse function F^{-1}
2) For q between 0 and 1, $F^{-1}(q)$ is called the qth quantile or 100qth percentile
3) Probability X is below its qth quantile is q: $P\{X \leqslant F^{-1}(q)\} = q$
   - 0.025 quantile = q(0.025) = 2.5% percentile = -1.96
   - 0.975 quantile = q(0.975) = 97.5% percentie = 1.96
   - Median = 0.5 quantile = q(0.5) = 50% percentie = 0
   - First quartiles = Q1 = 0.25 quantile = q(0.25) = 25% percentile
   - Third quartiles = Q3 = 0.75 quantile = q(0.75) = 75% percentile
   - For 95% confidence intervals we use the 0.025 and 0.975 quantiles

真题演练

Q2. What is a quantile? What is the value of the quantile at probability 0.1 (10%) of a standard normal? In which tail (left or right) of the distribution does it belong?

知识点梳理

4. Covariance & Correlation

1) Covariance σ_{xy}
   - $\sigma_{xy} = 0$, implies no linear relationship
   - does not imply independence, there could be a strong nonlinear association.

2) Correlation $\rho_{xy} = \sigma_{xy}/\sigma_x\sigma_y$

- ρ_{xy} between -1 and 1.
- An absolute correlation of 1 implies a linear relationship
- A strong nonlinear relationship may or may not imply a high correlation

Lecture 1.2 Returns: price and returns

知识点梳理

1. Price of an asset at time t (P_t)

2. Net Return

- $R_t = \frac{P_t}{P_{t-1}} - 1 = \frac{P_t - P_{t-1}}{P_{t-1}}$

真题演练

Q3. If P_{t-1} = 2 and P_t = 2.1 then find the net return.

知识点梳理

3. Simple gross return

- $\frac{P_t}{P_{t-1}} = 1 + R_t$

4. <u>Gross return over k periods (t−k to t)</u>

- $1 + R_t(k) = \frac{P_t}{P_{t-k}} = \left(\frac{P_t}{P_{t-1}}\right)\left(\frac{P_{t-1}}{P_{t-2}}\right) \dots \left(\frac{P_{t-k+1}}{P_{t-k}}\right)$

$= (1 + R_t) \dots (1 + R_{t-k+1})$

真题演练

Q4.

Time	$t-2$	$t-1$	t	$t+1$
P	200	210	206	212
$1+R$		1.05	.981	1.03
$1+R(2)$			1.03	1.01
$1+R(3)$				1.06

知识点梳理

5. Log returns

1) Log prices: $p_t = \log(P_t)$

2) Log returns

- $r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right) = p_t - p_{t-1}$
- Log returns are approximately equal to net returns

3) <u>**Advantage - Simplicity of multiperiod returns**</u>

- $r_t(k) = \log\big(1 + R_t(k)\big) = \log\big((1 + R_t) \dots (1 + R_{t-k+1})\big)$

  $= \log(1 + R_t) + \cdots + \log(1 + R_{t-k+1})$

- $\log\big(1 + R_t(k)\big) = r_t + r_{t-1} + \cdots + r_{t-k+1}$

➢ $r_t(2) = r_t + r_{t-1}$

➢ $r_t(3) = r_t + r_{t-1} + r_{t-2}$

➢ $r_t(4) = r_t + r_{t-1} + r_{t-2} + r_{t-3}$

真题演练

Q5. Suppose P_{t-1} = 2.0 and P_t = 2.06. Find R_t and r_t

Q6. Suppose P_{t-1} = 2.0 and P_t = 2.66. Find R_t and r_t

知识点梳理

6. Uncertainty in returns

1) Cross-sectional
   - Same time, different individual
2) Time series
   - Same individual, different time
3) Panel
   - Combine cross section and time series

7. IID Normal Returns

1) IID - $R_1, R_2, \ldots$ Independent and identically distributed
2) Two problems
   - Unlimited losses: $R_t \geq -1$ since you can lose no more than your investment
   - Not normal: $1 + R_t(k) = (1 + R_t) \ldots (1 + R_{t-k+1})$, product of normal variable is not normal.

8. The Lognormal Model

1) Assumes $r_t = \log(1+R_t)$ are IID and normal

2) Sums of normals are normal

真题演练

Q7. Asumme $r_t \sim N(0,1), r_{t-1} \sim N(0,1),\ r_{t-2} \sim N(0,1),\ r_{t-3} \sim N(0,1)$, all indepednet of one another. What is the density of $r_t(4)$?

Q8. $r_t \sim N(0,(0.1)^2)$, what is P(1 + R < 0.9)?

Q9. Suppose 1+R is lognormal (0, 0.25), what is P(1 + R < 1.3)?

Q10. Assume again that 1 + R is lognormal $\{0,(0.1)^2\}$, Find the probability that a simple gross two-period return is less than 0.9

Q11. Assume again that 1 + R is lognormal $\{0,(0.1)^2\}$, Find the probability that a simple gross three-period return is less than 0.9

Q12. Compute the probability that a simple gross three-period return is less than 0.7 given that gross returns are lognormally distributed with mean 0 and variance 4.

知识点梳理

9. Random Walk

1) Let $Z_1, Z_2, \ldots$ be IID with mean μ and standard deviation σ.

Z_0 is an arbitrary starting point. Let $S_0 = Z_0$; $S_t = Z_0 + Z_1 + \cdots + Z_t, t \geq 1$

$S_0, S_1, \ldots$ is called a random walk.

- $E(S_t|Z_0) = Z_0 + \mu t$
- $Var(S_t|Z_0) = \sigma^2 t$

- $SD(S_t|Z_0) = \sigma\sqrt{t}$

真题演练

Q12. What is the Random Walk? What do we know about its mean and variance?

知识点梳理

10. Geometric Random Walk

- Recall: $\log\big(1+R_t(k)\big) = r_t + r_{t-1} + \cdots + r_{t-k+1}$
- Therefore: $\log\left(\frac{P_t}{P_{t-k}}\right) = \log\big(1+R_t(k)\big) = \log P_t - \log P_{t-k}$
- Taking k = t

$\log\big(1+R_t(t)\big) = \log P_t - \log P_0$

$\log P_t = \log P_0 + r_1 + \cdots + r_t$

$\log P_t = \log P_{t-1} + r_t$

真题演练

Q13. $r_t \sim N(0,1)$ $r_1 = 1.336$; $r_2 = -0.484$; $r_3 = 1.628$; $r_4 = 1.508$; $\log P_0 = 0.1$. Find $\log P_1$, $\log P_2$, $\log P_3$, $\log P_4$

Q14. Suppose that log P_0 is known. Write the value of log P_3 as a function of log P_0 and returns r_t, t=1,2,3. Write the value of log P_3 as a function of log P_2 and r_3.

知识点梳理

11. Random walk with drift vs. Random walk

1) Random Walk with drift $r_t \sim N(\mu, \sigma^2)$
2) Random Walk $r_t \sim N(0, \sigma^2)$

- $S_t = Z_0 + \mu t + r_1 \ldots + r_t$ where $r_t \sim N(\mu, \sigma^2)$
- $\log P_t = \log P_0 + t\ constant + r_1 + \cdots + r_t$

真题演练

Q15. Compare the Random Walk and the Random Walk with drift

Q16. Are log-prices stationary? Are returns r_t stationary?

知识点梳理

12. Are log returns really normally distributed?

1) Normal probability plot (QQ Plot)
   - If approximately a straight line
2) Sample skewness and kurtosis
   - If near values of the normal distribution (skewness = 0, kurtosis = 3)
   - Skewness: measure of symmetry
     - $\hat{S} = \frac{1}{T}\sum_{t=1}^{T}(\frac{r_t-\hat{\mu}}{\hat{\sigma}})^3$
   - Kurtosis: measure of thickness (fat of tails)
     - $\hat{K} = \frac{1}{T}\sum_{t=1}^{T}(\frac{r_t-\hat{\mu}}{\hat{\sigma}})^4$
   - Excess kurtosis: measures deviation from 3, the kurtosis of a normal distribution
     - $\hat{K} - 3$
3) Three common tests of normality
   - Anderson-Darling test
   - Shapiro-Wilks test
   - Kolmogorov-Smirnov test
4) Hypothesis Test
   - H_0: return are normally distributed H_A: return are not normally distributed
   - P – value: probability of an error we allow for is 5%

     P – value < 0.05 reject H_0

     P – value ≥ 0.05 accept H_0

真题演练

Q17. What is skewness? What does it mean to investors that log-returns r_t are asymmetric?

Q18. What is kurtosis? What does it mean to investors that returns r_t are leptokurtic, i.e. have heavy(fat) tails?

Q19. Give the names of three tests that can be used to test the normality of returns. Write the null and alternative hypotheses of these tests.

Q20. If you have a sample of returns and the p-value of a Kolmogorov-Smirnov test is 0.005, what do you conclude?

知识点梳理

13. Efficient Market Hypothesis

1) A market is "efficient with respect to an information set" if prices would be unchanged by revealing that information to all participants

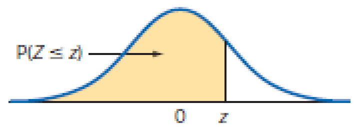
真题演练

Q21. What is the Efficient Market Hypothesis? What models of log-prices does it suggest?

Q22. Are log-prices predictable given the observations on their past?

Table III Standard Normal Distribution Cumulative Probabilities

Let Z be a standard normal random variable: $\mu = 0$ and $\sigma = 1$.
This table contains cumulative probabilities: $P(Z \le z)$.



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table III Standard Normal Distribution Cumulative Probabilities (*Continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Special critical values: $P(Z \geq z_\alpha) = \alpha$

α	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
z_α	1.2816	1.6449	1.9600	2.3263	2 .5758	3.0902	3.2905	3.7190

α	0.00009	0.00008	0.00007	0.00006	0.00005	0.00004	0.00003	0.00002	0.00001
z_α	3.7455	3.7750	3.8082	3.8461	3.8906	3.9444	4.0128	4.1075	4.2649